



Professional Educational Materials

Student Number

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2023 PRELIMINARY EXAMINATION

Mathematics Standard

General Instructions

- Reading time – 5 minutes
- Working time - 2 hours
- Write using black pen
- NESA approved calculators may be used
- A NESA reference sheet is provided
- In Questions 13 – 30, show relevant mathematical reasoning and/or calculations

Total Marks – 80

Section I Questions 1 – 12 **12 marks**

Allow about 20 minutes for this section

Section II Questions 13 – 30 **68 marks**

Allow about 1 hour and 40 minutes for this section

Directions to School or College

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Section I (12 marks)

Write your answers on the multiple-choice answer sheet.

Allow about 20 minutes for this section.

- 1 Which answer represents the number 145 086 in standard notation correct to two significant figures?

(A) 15×10^4
(B) 145×10^5
(C) 1.4×10^5
(D) 1.5×10^5

- 2 Which set of data is classified as categorical and nominal?

(A) small, medium, large
(B) blue, red, purple
(C) 1, 2, 3, 4, 5
(D) 5 cats, 3 cats, 1 cat

- 3 Evaluate the following equation for d

$$d = \sqrt{u^2 + 2as} \text{ by substituting } u = 4, a = 5 \text{ and } s = 10.$$

(A) 10
(B) 116
(C) 10.8
(D) 9.2

- 4 A machine operator earns a yearly salary of \$56 200 before tax.
How much does he earn per fortnight?

(A) \$1080.77
(B) \$4014.29
(C) \$2161.54
(D) \$5620.00

- 5 Convert 90 km/h to m/s.

(A) 25m/s
(B) 90000m/s
(C) 1.5m/s
(D) 259m/s

- 6 A 200g box of chocolate chip cookies has the following nutritional information.

	Average Quantity per serve Approx. 4 cookies (50 g)	Average Quantity per 100 g
Energy	760 kj	1520 kj
Protein	15.8 g	31.6 g
Fat	6.9 g	13.7 g
Carbohydrate	2.4 g	4.7 g
Sodium	221 mg	442 mg
Dietary fibre	6.4g	12.8g

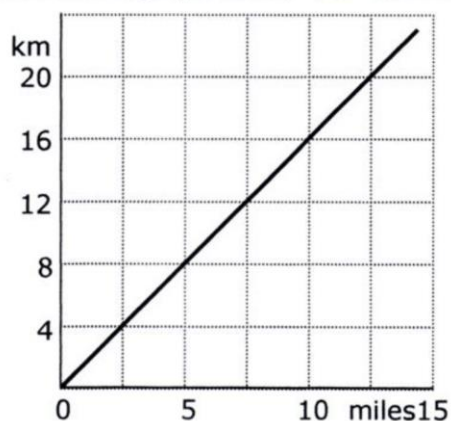
If Mari has 2 cookies a day and the recommended daily intake for an adult is 8700 kj per day.

What percentage of the recommended daily intake does she have left for other food?

(A) 91.3%
(B) 4.37%
(C) 95.6%
(D) 8.74%

- 7 The graph to convert miles to kilometres is shown below.

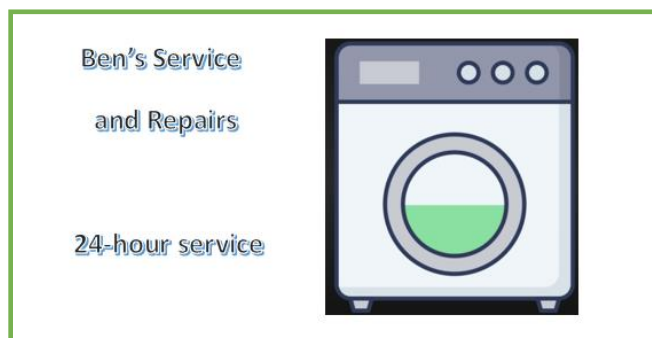
Conversion graph miles $\longleftrightarrow$ kilometres



The relationship between x and y is?

- (A) $y = \frac{8}{5}x$
- (B) $y = \frac{5}{8}x$
- (C) $y = 2x$
- (D) $y = x$

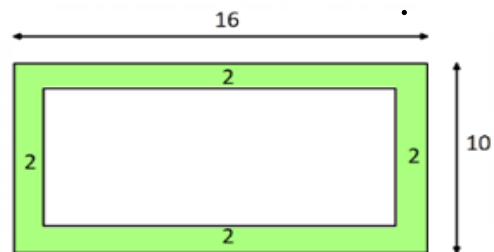
- 8 A serviceman charges \$79.00 including 10% GST for a call-out.



What is the cost of a call-out before GST is added?

- (A) \$7.90
- (B) \$86.90
- (C) \$71.10
- (D) \$71.82

- 9 Below is the design for a picture frame. All measurements are in centimetres and the width of the frame is 2 cm.



Calculate the shaded area in the diagram shown above.

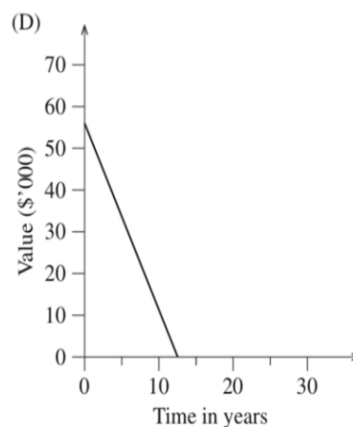
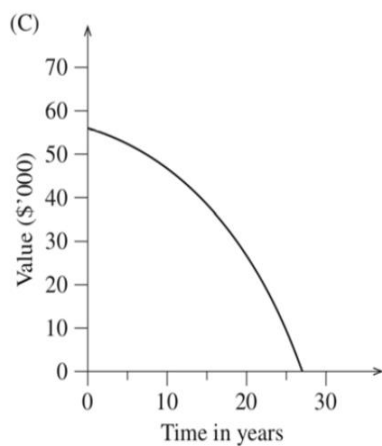
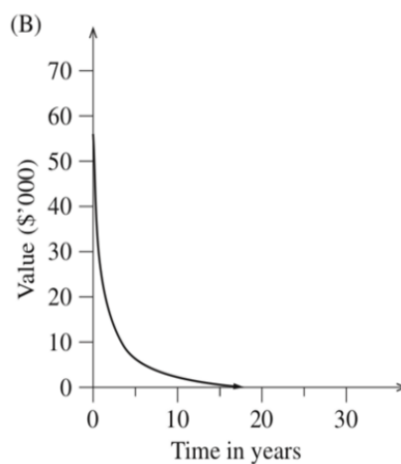
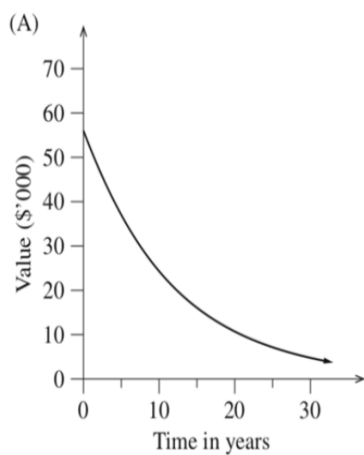
- (A) 160cm^2
 (B) 88cm^2
 (C) 156m^2
 (D) 48cm^2
- 10 Ian is male and weighs 72kg. He holds a provisional licence with a zero-alcohol limit. He started drinking at 6 pm and had 10 schooners of full-strength beer (15 standard drinks) over 6 hours. He stopped drinking at midnight. What is his BAC at midnight?

$$BAC_{female} = \frac{10N - 7.5H}{5.5M} \quad BAC_{male} = \frac{10N - 7.5H}{6.8M},$$

where N is the number of standard drinks consumed, H is the number of hours drinking and M is the person's weight in kilograms.

- (A) 0.21
 (B) 0.27
 (C) 0.11
 (D) 0.14

- 11 A new hybrid car, initially worth \$56000, depreciates at 8% per annum. Which graph best shows the salvage value of this car over time?



- 12 A die with 15 faces marked 1 to 15 is rolled. What is the probability that the uppermost face is divisible by 3?

- (A) $\frac{1}{5}$
 (B) $\frac{1}{3}$
 (C) $\frac{4}{15}$
 (D) $\frac{1}{15}$

END OF SECTION I

Student Number

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Section II (68 marks)

Attempt Questions 13 - 30

Allow about 1 hour and 40 minutes for this section.

Answer the questions in the spaces provided.

Your responses should include relevant mathematical reasoning and/or calculations.

Extra writing space is provided at the end of the section. If you use this space, clearly indicate which question you are answering.

Write your Student Number at the top of this page.

Please turn over

Question 13. (2 marks)

Solve the equation $5(1-x) = 3(x-1)$.

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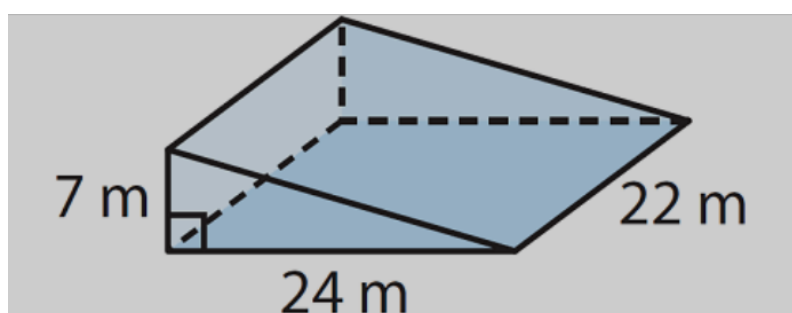
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Question 14. (5 marks)

The skateboard ramp pictured below needs to be filled with water for stability.



- a) Find the volume of water needed to fill the ramp. (Give your answer in litres)

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- b) How much will it cost, to the nearest dollar, to paint the 2 triangular sides and top (the back and base do not need painting) of the ramp if paint costs \$12 per square metre?

2

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Question 15. (3 marks)

Hung purchased a new car for \$43700 on the 23rd of January 2022.

- a) Calculate to 2 decimal places the percentage decrease in the price of the new car if the market value after 3 years is \$37000? **1**

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- b) What is the annual amount of depreciation if the amount of depreciation is constant? **1**

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- c) Hung has a no-claim bonus of 40% with MRMA. If the insurance premium is usually \$1690 calculate the cost of the premium for Hung if he claims his bonus? **1**

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Question 16. (1 mark)

Perform the following calculation and round your answer to 2 significant figures. **1**

$$52.79 \times (2.3 \times 10^3)$$

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Question 17. (2 marks)

Fried's rule calculates the dose (D) of medication for a child that is 1 – 2 years using the age of the child in months (m), where A is the adult dose in mg. **2**

$$D = \frac{mA}{140}$$

Calculate to 1 decimal place the dosage in mg for a one-and-a-half-year-old child weighing 11 kg if the adult dosage is 400 mg.

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Question 18. (4 marks)

Consider the linear equation $2x + 3y - 1 = 0$

- a) Find the gradient. 1

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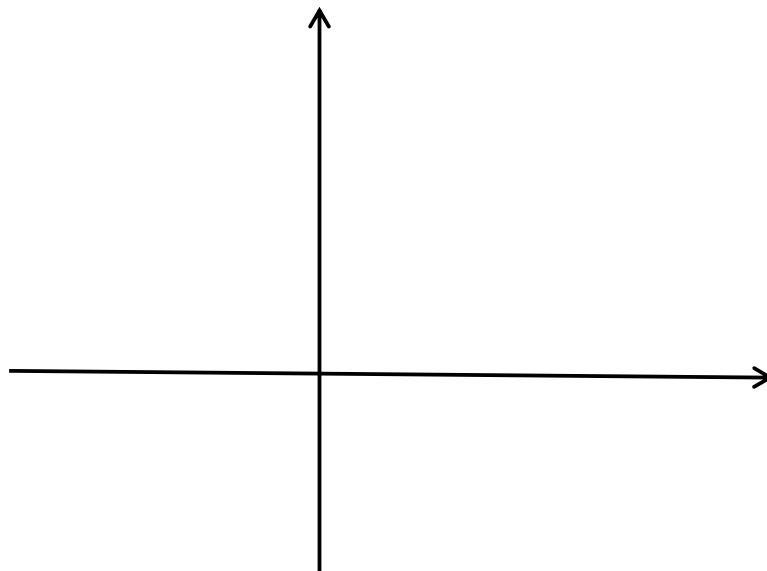
- b) Find the x -intercept. 1

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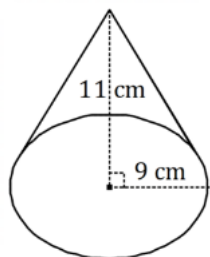
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- c) Sketch the graph of the function. 2



Question 19. (4 marks)

Gina is making cones to hold hot chips at the easter show. She is given this diagram as a guide.



- a) Use the formula for the surface area of a cone $SA = \pi r^2 + \pi rl$, where l is the slant side, to find the surface area of one cone. Give your answer to two decimal places. 2

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- b) The dimensions of a sheet of cardboard are 70cm x 90cm. 1
 If she fits 6 cones on one sheet, how many sheets of cardboard will she need to make 120 cones.

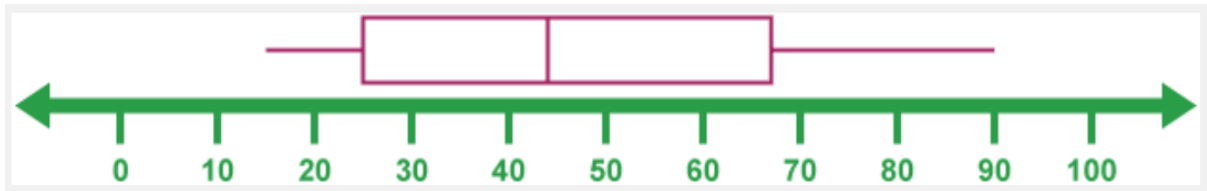
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- c) How much cardboard will be left over from each sheet? 1

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Question 20. (5 marks)

Consider the following box and whisker plot.



- a) Describe the data in terms of shape, centre and spread? 2

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- b) Find the range and interquartile range of the data? 2

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- a) What is the median of the data? 1

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Question 21. (4 marks)

Bennett has a packet of lollies. His favourite ones are the red ones and the purple ones.

The table below shows the probability of each different coloured lolly being selected. The probability of selecting a yellow lolly has been left out.

Lolly colour	Purple	Red	Orange	Green	Yellow
Probability	0.15	0.2	0.3	0.16	

- a) What is the probability that he chooses a yellow lolly? 2

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- b) There are 35 lollies in a packet. 2

How many of these lollies are expected to be his favourites?

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Question 22. (3 marks)

A private pool is charged a peak rate of 26.334 cents per kWh and an off-peak rate of 12.199 cents per kWh (for usage between 10 pm and 7am).

They use their 2000 W pump for two and a half hours per day, after 10pm, six days a week when the pool closes. How much will their yearly bill be for the pool alone?

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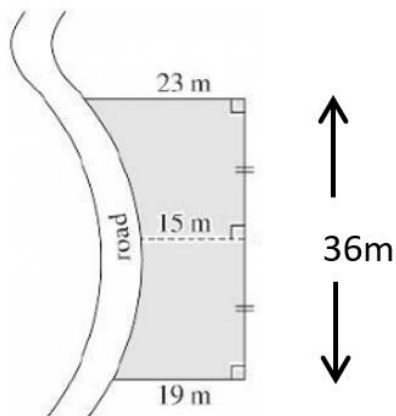
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Question 23. (5 marks)

- a) Use two applications of the Trapezoidal rule to estimate the area of the plot of land shown in the diagram below.

2



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- b) If the land has to be dug out to a depth of 200mm, find the volume of dirt removed? Answer in cubic meters.

3

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Question 24. (3 marks)

Consider the data presented below in a stem and leaf plot.

The plot shows the time in minutes to complete a decathlon.

Stem	Leaf
9	1 5 7 9
10	2 4 5 6 6 8
11	0 2 2 3 4 4 5
12	1 2 3 3 3 7 9
13	2 3 4 5 7 7
14	3 6 9
15	0 1 2

- a) How many contestants participated in the decathlon? **1**

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- b) What was the median time in minutes to finish the race? **1**

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- c) Contestants with a time under 125 minutes are entered into the next round. **1**
What percentage of contestants made it to the next round?

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Question 25. (4 marks)

A Hybrid solar eclipse was seen from Exmouth in Western Australia on 20th April 2023.

If the partial eclipse began at 10:04am and the total eclipse went from 11:48am to 11:49am then partial eclipse again from 11:49am to 1:23pm.



- a) What was the total time where the sun was in partial eclipse? 2

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- b) Sydney is 2 hours ahead of Western Australia. 2

Give the start and finish times for someone watching the **total** eclipse on TV in Sydney.

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Question 26. (4 marks)

Ollie has a gross annual salary of \$68 500.

He has yearly tax deductions of \$750 for work related travel and \$200 for uniforms.

The Medicare levy that he pays is calculated at 2% of his taxable income.

Ollie has already paid \$15400 in tax. Will Ollie receive a tax refund, or will

he owe money to the Australian Taxation Office? Justify your answer by

calculating the refund or amount owed using the Tax Table below.

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Taxable income	Tax on this income
0 – \$18,200	Nil
\$18,201 – \$45,000	19 cents for each \$1 over \$18,200
\$45,001 – \$120,000	\$5,092 plus 32.5 cents for each \$1 over \$45,000
\$120,001 – \$180,000	\$29,467 plus 37 cents for each \$1 over \$120,000
\$180,001 and over	\$51,667 plus 45 cents for each \$1 over \$180,000

The above rates do not include the Medicare levy of 2%.

Question 27. (4 marks)

Australia plays India in a cricket test series comprising of 3 games.

There are 3 possible outcomes for each game win, draw or lose.

Based on previous game data the probability of Australia winning a game is 0.3 and of losing is 0.4. Australia wins the first game.

- a) Draw a tree diagram and list the possible outcomes of the other 2 games. 2

- b) To win the series, Australia needs to win at least 2 games out of the three. 2

Find the probability that Australia wins the series?

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Question 28. (6 marks)

The following table shows the time in minutes that it took some students to complete an obstacle course at their Year 11 camp.

Time, t (minutes)	Frequency
$0 < t \leq 10$	8
$10 < t \leq 20$	21
$20 < t \leq 30$	17
$30 < t \leq 40$	5
$40 < t \leq 50$	2

- a) How many students competed in the obstacle course? 1

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- b) Find the modal class. 1

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- c) What percentage of students took more than 20 minutes to complete the course? 2

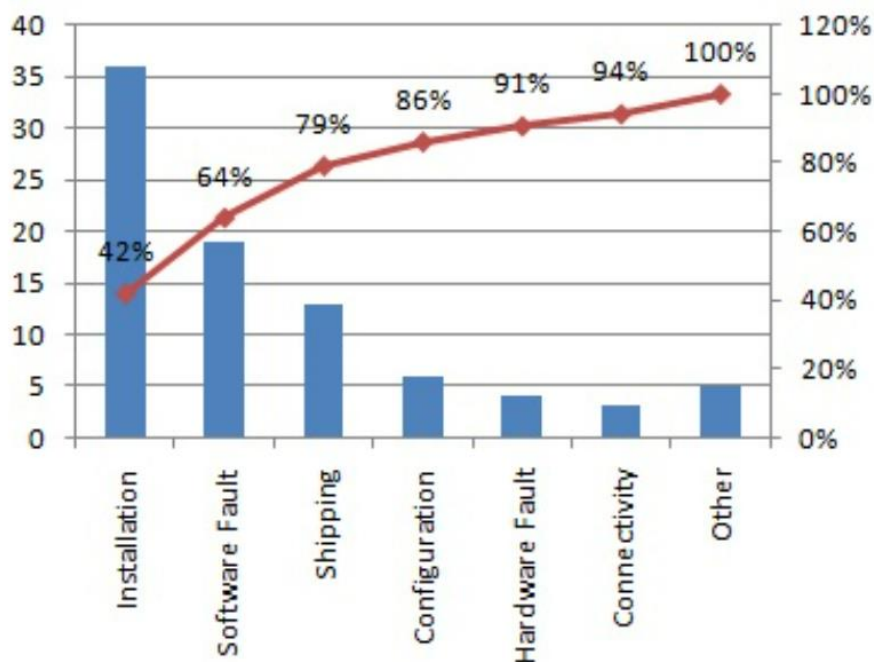
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- d) Find the class containing the median 2

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Question 29. (3 marks)

An online business organiser asked customers to fill in a survey. The feedback was organised into the following Pareto chart.



- a) What is the approximate cumulative percentage for Shipping?

..... 1

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- b) Which two most common complaints should the business work on to improve customer satisfaction? 2

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Question 30. (6 marks)

A sum of \$50000 is invested into an account when Fred retires at 67 years old. It is to pay for his retirement. Fred estimates he will live until 90 years old. The savings account attracts 3% per year simple interest.

- a) Calculate the amount of interest earned each year. **1**

.....

- b) Complete the table below to calculate the amount the investment is worth over the next 6 years. **3**

Age	Amount in account
67	50000
68	51500
69	
70	
71	
72	
73	

How much will he have in the account in 6 years' time?

.....

- c) At age 73 Fred buys a car for \$32000.

He pays a deposit of \$20000 from his account. Fred agrees to pay the car finance company back the balance at 7% simple interest for 10 years.

2

How much simple interest will he pay on the loan to the car finance company?

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Section II extra writing space

If you use this space, clearly indicate which question you are answering.

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Section II extra writing space

If you use this space, clearly indicate which question you are answering.

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REFERENCE SHEET

Measurement

Precision

$$\text{Absolute error} = \frac{1}{2} \times \text{precision}$$

$$\text{Upper bound} = \text{measurement} + \text{absolute error}$$

$$\text{Lower bound} = \text{measurement} - \text{absolute error}$$

Length, area, surface area and volume

$$l = \frac{\theta}{360} \times 2\pi r$$

$$A = \frac{\theta}{360} \times \pi r^2$$

$$A = \frac{h}{2}(x + y)$$

$$A \approx \frac{h}{2}(d_f + d_l)$$

$$A = 2\pi r^2 + 2\pi rh$$

$$A = 4\pi r^2$$

$$V = \frac{1}{3}Ah$$

$$V = \frac{4}{3}\pi r^3$$

Trigonometry

$$A = \frac{1}{2}ab \sin C$$

$$\frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C}$$

$$c^2 = a^2 + b^2 - 2ab \cos C$$

$$\cos C = \frac{a^2 + b^2 - c^2}{2ab}$$

Financial Mathematics

$$FV = PV(1 + r)^n$$

Straight-line method of depreciation

$$S = V_0 - Dn$$

Declining-balance method of depreciation

$$S = V_0(1 - r)^n$$

Statistical Analysis

$$z = \frac{x - \bar{x}}{s}$$

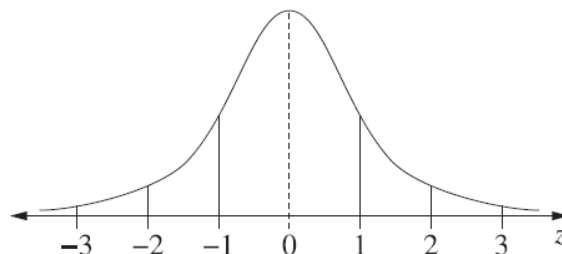
An outlier is a score

less than $Q_1 - 1.5 \times IQR$

or

more than $Q_3 + 1.5 \times IQR$

Normal distribution



- approximately 68% of scores have z-scores between -1 and 1
- approximately 95% of scores have z-scores between -2 and 2
- approximately 99.7% of scores have z-scores between -3 and 3

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MULTIPLE CHOICE ANSWER SHEET (12 marks)

Name:

Student Number:

Choose the best answer by filling in the circle corresponding to *A*, *B*, *C* or *D*.

	<i>A</i>	<i>B</i>	<i>C</i>	<i>D</i>
1	☐	☐	☐	☐
2	☐	☐	☐	☐
3	☐	☐	☐	☐
4	☐	☐	☐	☐
5	☐	☐	☐	☐
6	☐	☐	☐	☐
7	☐	☐	☐	☐
8	☐	☐	☐	☐
9	☐	☐	☐	☐
10	☐	☐	☐	☐
11	☐	☐	☐	☐
12	☐	☐	☐	☐